Causal Analysis Report

bpic18 Parcel document.xes.gz-im

Decision points:	260453
Choice sets:	14
Transitions fitted:	6
Significant (p<0.05):	0
Generated:	2026-06-22 09:02

Step 1 — Association test (logistic regression + permutation)
H_0: choice ⊥ util | choice_set.   reject_H0=True → util predicts routing.

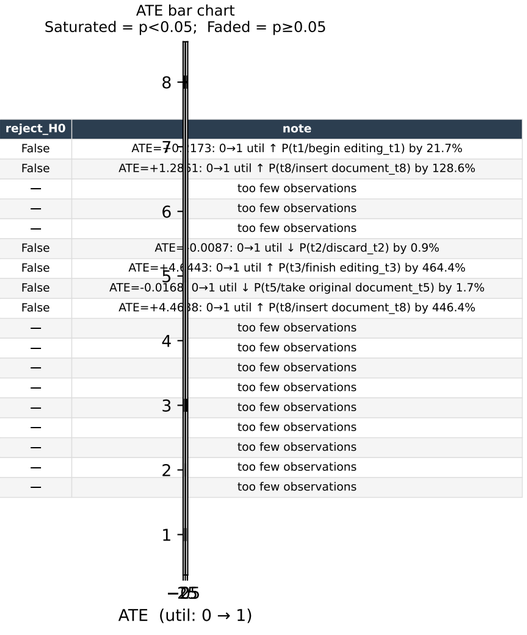
(no data)

Step 2 — Double ML (partially linear regression, HC3 robust SE)
θ = linear effect of 1-unit ↑ util on P(chose k), after partialling out X.
Equals ATE when the effect is homogeneous. Otherwise: variance-weighted average of CATEs.

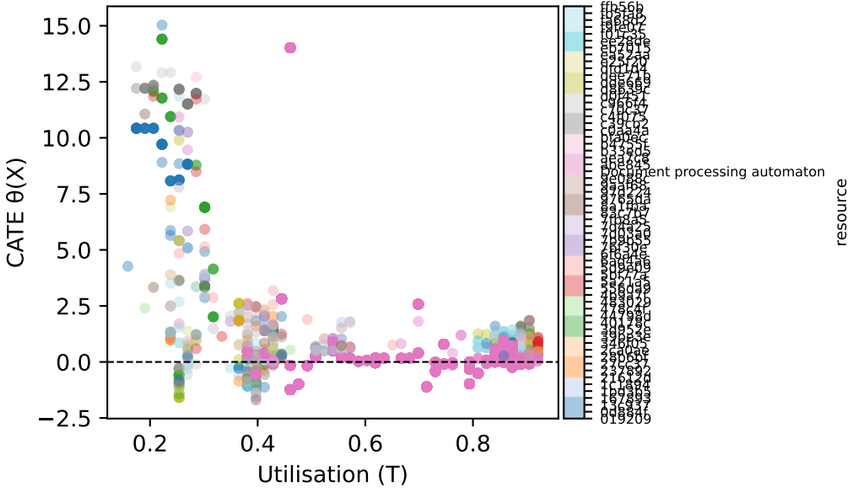
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	theta	se	ci_95	p_value	reject_H0	note
(1, 2, 3, 4, 5, 8)	('begin editing_t1', 'discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	14750	1, 8	1	begin editing_t1	discard_t2, finish editing_t3, insert document_t8, take original document_t5	0.131183	0.061959	[0.0097, 0.2526]	0.0342	True	θ=+0.1312: 1-unit ↑ util ↑ P(t1/begin editing_t1) by 13.1%
(1, 2, 3, 4, 5, 8)	('begin editing_t1', 'discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	14750	1, 8	8	insert document_t8	begin editing_t1, discard_t2, finish editing_t3, take original document_t5	0.444791	0.157571	[0.1360, 0.7536]	0.0048	True	θ=+0.4448: 1-unit ↑ util ↑ P(t8/insert document_t8) by 44.5%
(1, 9, 10, 11, 12)	('begin editing_t1', 'τ_t9', 'τ_t10', 'τ_t11', 'remove document_t12')	15	1	—	—	—	nan	nan	—	nan	—	too few observations
(1, 13)	('begin editing_t1', 'save_t13')	186	13	—	—	—	nan	nan	—	nan	—	too few observations
(1, 14, 15)	('begin editing_t1', 'τ_t14', 'τ_t15')	186	1	—	—	—	nan	nan	—	nan	—	too few observations
(2, 3, 4, 5, 8)	('discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	55555	2, 3, 5, 8	2	discard_t2	finish editing_t3, insert document_t8, take original document_t5	-0.002173	0.001813	[-0.0057, 0.0014]	0.2305	False	θ=-0.0022: 1-unit ↑ util ↓ P(t2/discard_t2) by 0.2%
(2, 3, 4, 5, 8)	('discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	55555	2, 3, 5, 8	3	finish editing_t3	discard_t2, insert document_t8, take original document_t5	6.115782	0.099732	[5.9203, 6.3113]	0.0	True	θ=+6.1158: 1-unit ↑ util ↑ P(t3/finish editing_t3) by 611.6%
(2, 3, 4, 5, 8)	('discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	55555	2, 3, 5, 8	5	take original document_t5	discard_t2, finish editing_t3, insert document_t8	-0.000434	0.000225	[-0.0009, 0.0000]	0.0543	False	θ=-0.0004: 1-unit ↑ util ↓ P(t5/take original document_t5) by 0.0%
(2, 3, 4, 5, 8)	('discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	55555	2, 3, 5, 8	8	insert document_t8	discard_t2, finish editing_t3, take original document_t5	5.474735	0.102103	[5.2746, 5.6749]	0.0	True	θ=+5.4747: 1-unit ↑ util ↑ P(t8/insert document_t8) by 547.5%
(6, 7)	('τ_t6', 'check_t7')	13588	7	—	—	—	nan	nan	—	nan	—	too few observations
(9, 10, 11, 12)	('τ_t9', 'τ_t10', 'τ_t11', 'remove document_t12')	2	12	—	—	—	nan	nan	—	nan	—	too few observations
(10, 11, 12)	('τ_t10', 'τ_t11', 'remove document_t12')	4	12	—	—	—	nan	nan	—	nan	—	too few observations
(11, 12)		0		—	—	—	nan	nan	—	nan	—	too few observations
(11, 12, 16, 17)	('τ_t11', 'remove document_t12', 'τ_t16', 'change department_t17')	2	12	—	—	—	nan	nan	—	nan	—	too few observations
(13, 16, 17)	('save_t13', 'τ_t16', 'change department_t17')	6629	13	—	—	—	nan	nan	—	nan	—	too few observations
(14, 15)		0		—	—	—	nan	nan	—	nan	—	too few observations
(14, 15, 16, 17)		0		—	—	—	nan	nan	—	nan	—	too few observations
(15, 16, 17)	('τ_t15', 'τ_t16', 'change department_t17')	5	17	—	—	—	nan	nan	—	nan	—	too few observations

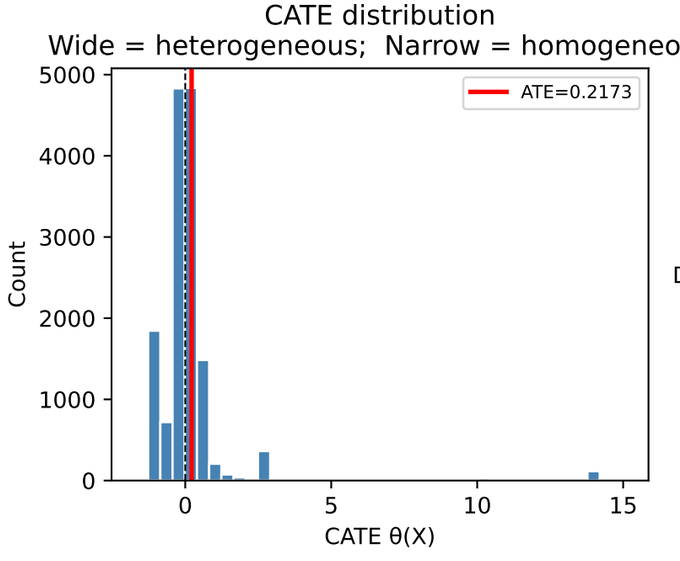
Step 3 — Causal Forest DML (ATE per transition)
ATE = mean of per-sample CATEs. Green rows = H0 rejected (p<0.05).
Less powerful for significance than Double ML, but basis for heterogeneity analysis.

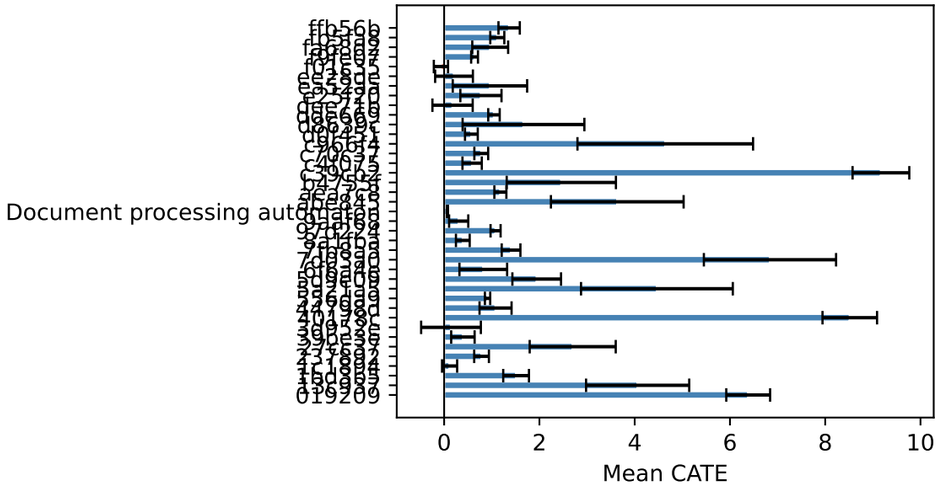
choice_set	choice_set_labels	n	choices_seen	transition	transition_label	alt_utils	ate	ci_95	p_value	reject_H0	note
(1, 2, 3, 4, 5, 8)	('begin editing_t1', 'discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	14750	1, 8	1	begin editing_t1	discard_t2, finish editing_t3, insert document_t8, take original document_t5	0.217271	[-4.0128, 4.4473]	0.9198	False	ATE=+0.2173: 0→1 util ↑ P(t1/begin editing_t1) by 21.7%
(1, 2, 3, 4, 5, 8)	('begin editing_t1', 'discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	14750	1, 8	8	insert document_t8	begin editing_t1, discard_t2, finish editing_t3, take original document_t5	1.286132	[-23.0818, 25.6540]	0.9176	False	ATE=+1.2861: 0→1 util ↑ P(t8/insert document_t8) by 128.6%
(1, 9, 10, 11, 12)	('begin editing_t1', 'τ_t9', 'τ_t10', 'τ_t11', 'remove document_t12')	15	1	—	—	—	nan	—	nan	—	too few observations
(1, 13)	('begin editing_t1', 'save_t13')	186	13	—	—	—	nan	—	nan	—	too few observations
(1, 14, 15)	('begin editing_t1', 'τ_t14', 'τ_t15')	186	1	—	—	—	nan	—	nan	—	too few observations
(2, 3, 4, 5, 8)	('discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	55555	2, 3, 5, 8	2	discard_t2	finish editing_t3, insert document_t8, take original document_t5	-0.008693	[-0.3042, 0.2868]	0.954	False	ATE=-0.0087: 0→1 util ↓ P(t2/discard_t2) by 0.9%
(2, 3, 4, 5, 8)	('discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	55555	2, 3, 5, 8	3	finish editing_t3	discard_t2, insert document_t8, take original document_t5	4.644344	[-5.2439, 14.5326]	0.3573	False	ATE=+4.6443: 0→1 util ↑ P(t3/finish editing_t3) by 464.4%
(2, 3, 4, 5, 8)	('discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	55555	2, 3, 5, 8	5	take original document_t5	discard_t2, finish editing_t3, insert document_t8	-0.016784	[-0.0168, -0.0168]	1.0	False	ATE=-0.0168: 0→1 util ↓ P(t5/take original document_t5) by 1.7%
(2, 3, 4, 5, 8)	('discard_t2', 'finish editing_t3', 'τ_t4', 'take original document_t5', 'insert document_t8')	55555	2, 3, 5, 8	8	insert document_t8	discard_t2, finish editing_t3, take original document_t5	4.463757	[-14.0039, 22.9314]	0.6357	False	ATE=+4.4638: 0→1 util ↑ P(t8/insert document_t8) by 446.4%
(6, 7)	('τ_t6', 'check_t7')	13588	7	—	—	—	nan	—	nan	—	too few observations
(9, 10, 11, 12)	('τ_t9', 'τ_t10', 'τ_t11', 'remove document_t12')	2	12	—	—	—	nan	—	nan	—	too few observations
(10, 11, 12)	('τ_t10', 'τ_t11', 'remove document_t12')	4	12	—	—	—	nan	—	nan	—	too few observations
(11, 12)		0		—	—	—	nan	—	nan	—	too few observations
(11, 12, 16, 17)	('τ_t11', 'remove document_t12', 'τ_t16', 'change department_t17')	2	12	—	—	—	nan	—	nan	—	too few observations
(13, 16, 17)	('save_t13', 'τ_t16', 'change department_t17')	6629	13	—	—	—	nan	—	nan	—	too few observations
(14, 15)		0		—	—	—	nan	—	nan	—	too few observations
(14, 15, 16, 17)		0		—	—	—	nan	—	nan	—	too few observations
(15, 16, 17)	('τ_t15', 'τ_t16', 'change department_t17')	5	17	—	—	—	nan	—	nan	—	too few observations

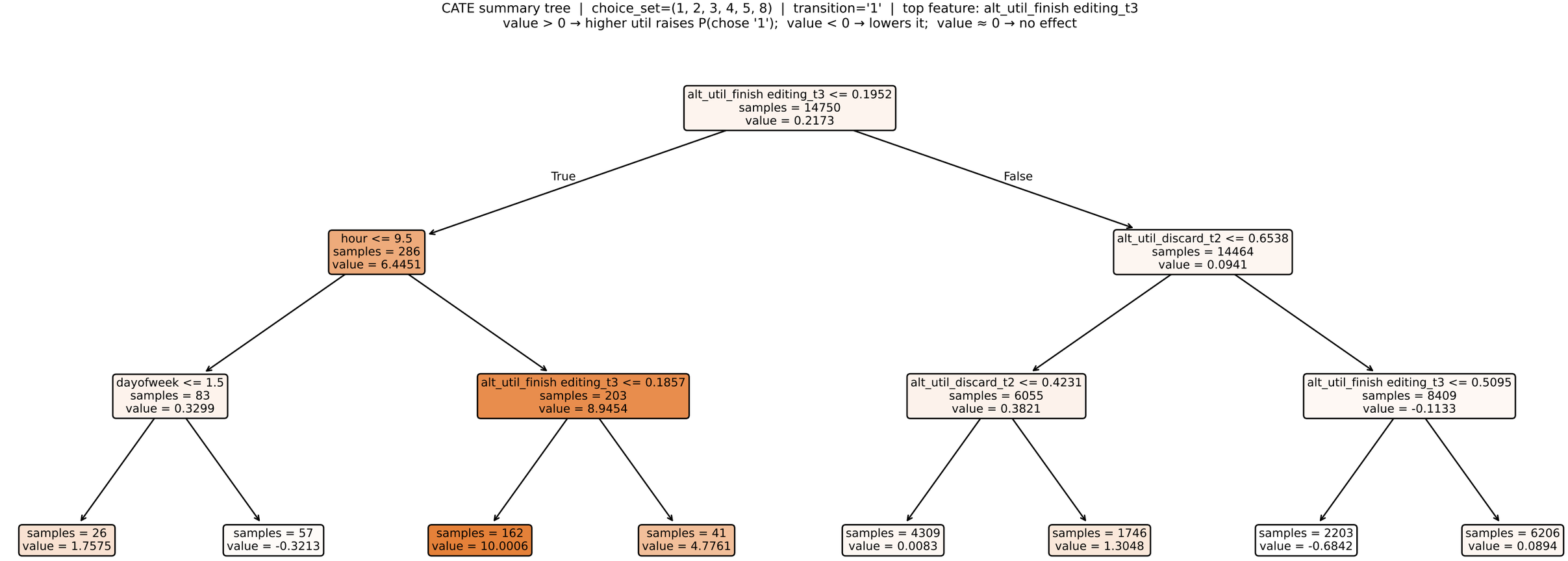


CATE — choice_set=(1, 2, 3, 4, 5, 8) | transition='1' | n=14750 | ATE=0.2173 std=1.7907

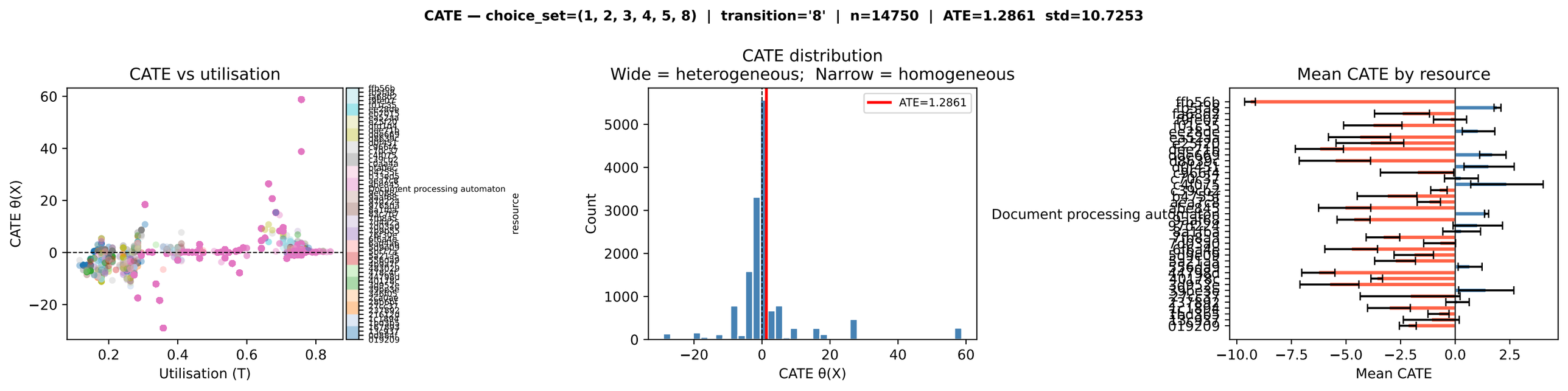


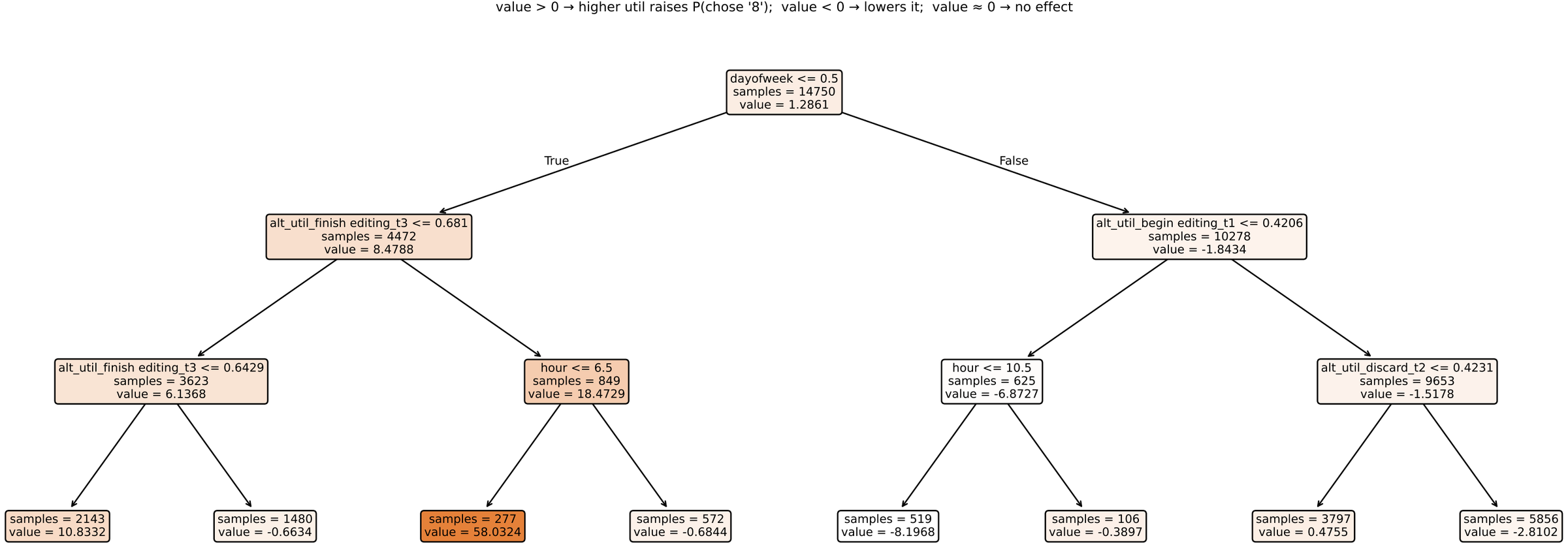


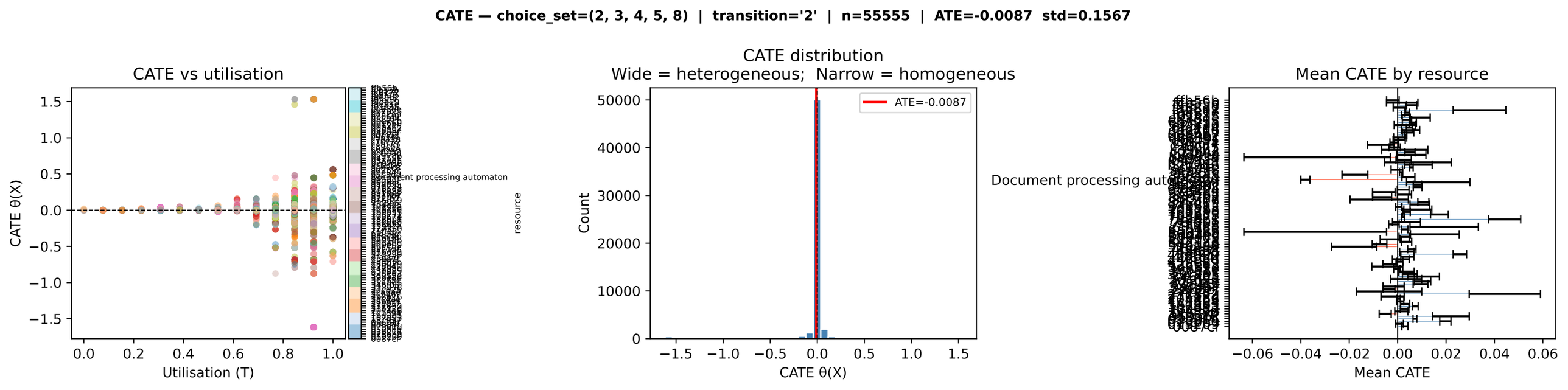


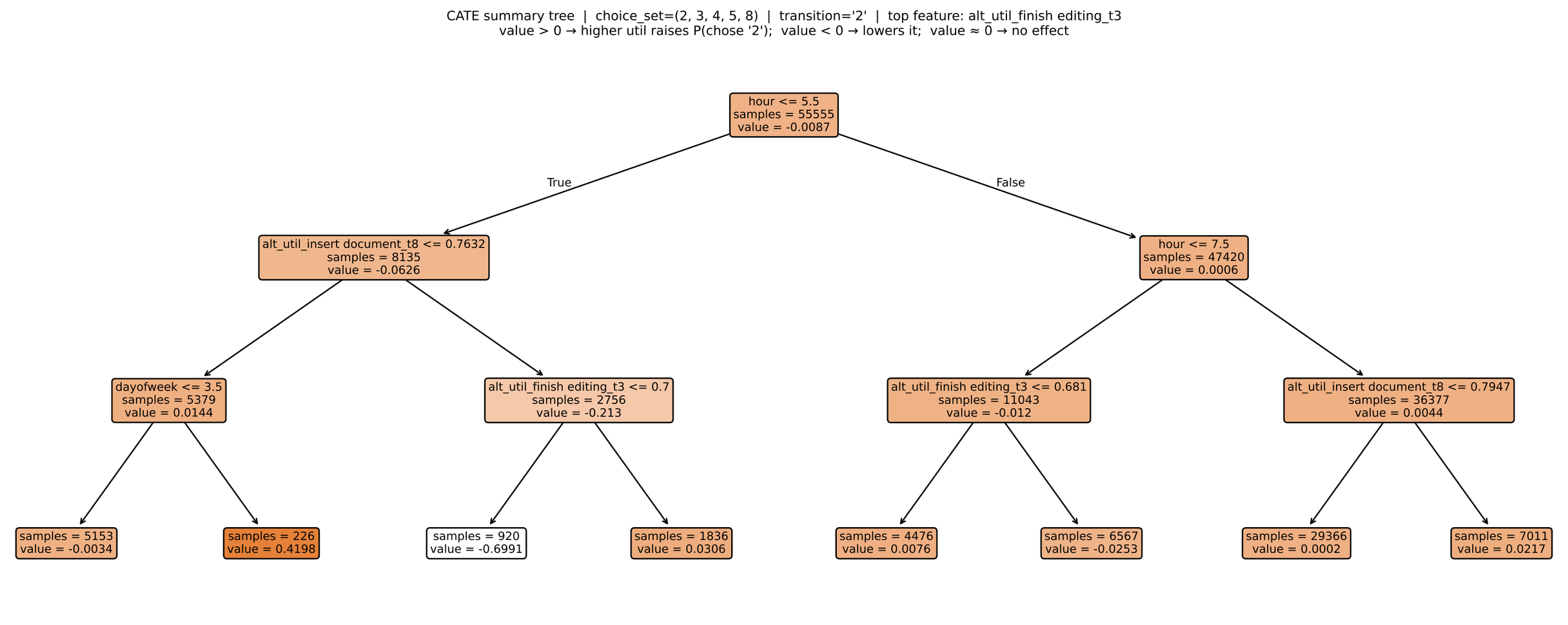

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 8) | transition='1' | top feature: alt_util_finish editing_t3
value > 0 → higher util raises P(chose '1'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_finish editing_t3 <= 0.1952
samples = 14750
value = 0.2173
True
False
hour <= 9.5
samples = 286
value = 6.4451
alt_util_discard_t2 <= 0.6538
samples = 14464
value = 0.0941
dayofweek <= 1.5
samples = 83
value = 0.3299
alt_util_finish editing_t3 <= 0.1857
samples = 203
value = 8.9454
alt_util_discard_t2 <= 0.4231
samples = 6055
value = 0.3821
alt_util_finish editing_t3 <= 0.5095
samples = 8409
value = -0.1133
samples = 26
value = 1.7575
samples = 57
value = -0.3213
samples = 162
value = 10.0006
samples = 41
value = 4.7761
samples = 4309
value = 0.0083
samples = 1746
value = 1.3048
samples = 2203
value = -0.6842
samples = 6206
value = 0.0894

CATE — choice_set=(1, 2, 3, 4, 5, 8) | transition='8' | n=14750 | ATE=1.2861 std=10.7253

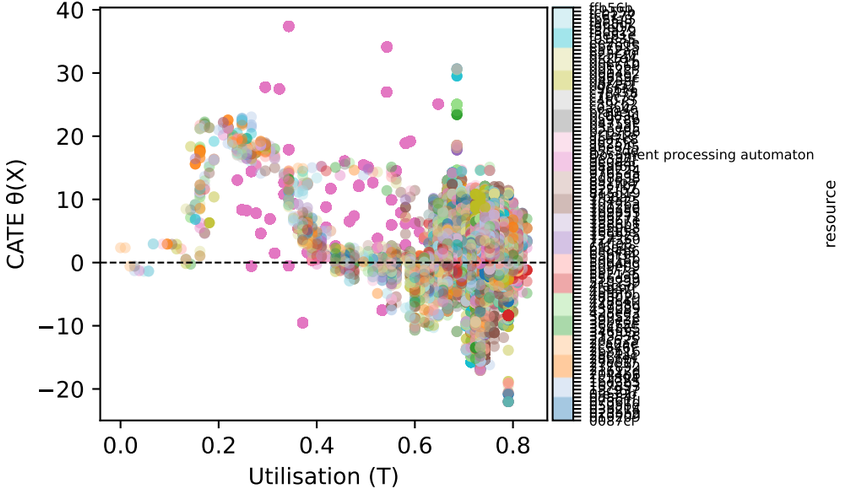


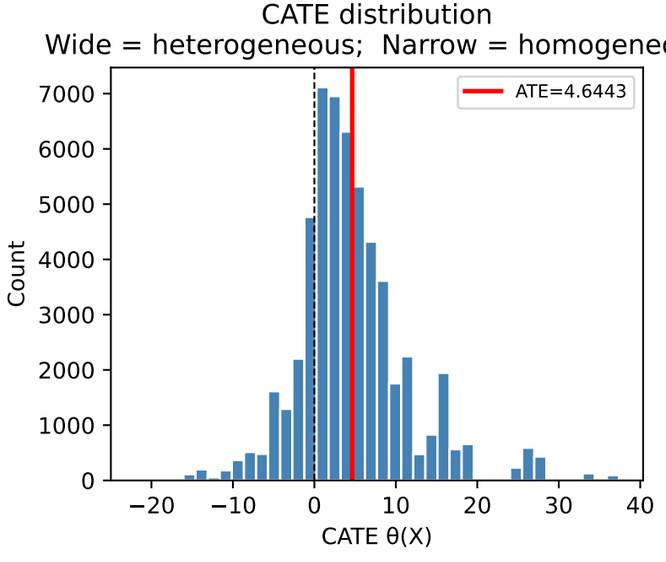

CATE summary tree | choice_set=(1, 2, 3, 4, 5, 8) | transition='8' | top feature: hour
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
dayofweek <= 0.5
samples = 14750
value = 1.2861
True
False
alt_util_finish editing_t3 <= 0.681
samples = 4472
value = 8.4788
alt_util_begin editing_t1 <= 0.4206
samples = 10278
value = -1.8434
alt_util_finish editing_t3 <= 0.6429
samples = 3623
value = 6.1368
hour <= 6.5
samples = 849
value = 18.4729
hour <= 10.5
samples = 625
value = -6.8727
alt_util_discard_t2 <= 0.4231
samples = 9653
value = -1.5178
samples = 2143
value = 10.8332
samples = 1480
value = -0.6634
samples = 277
value = 58.0324
samples = 572
value = -0.6844
samples = 519
value = -8.1968
samples = 106
value = -0.3897
samples = 3797
value = 0.4755
samples = 5856
value = -2.8102

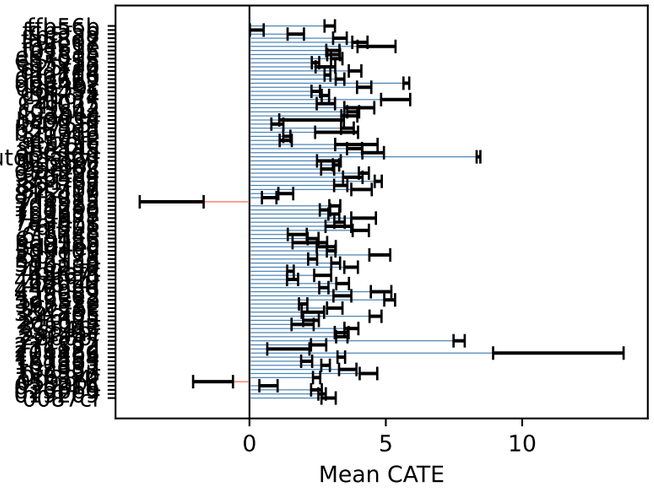

CATE — choice_set=(2, 3, 4, 5, 8) | transition='2' | n=55555 | ATE=-0.0087 std=0.1567
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0087
Count
CATE θ(X)
Mean CATE by resource
Mean CATE

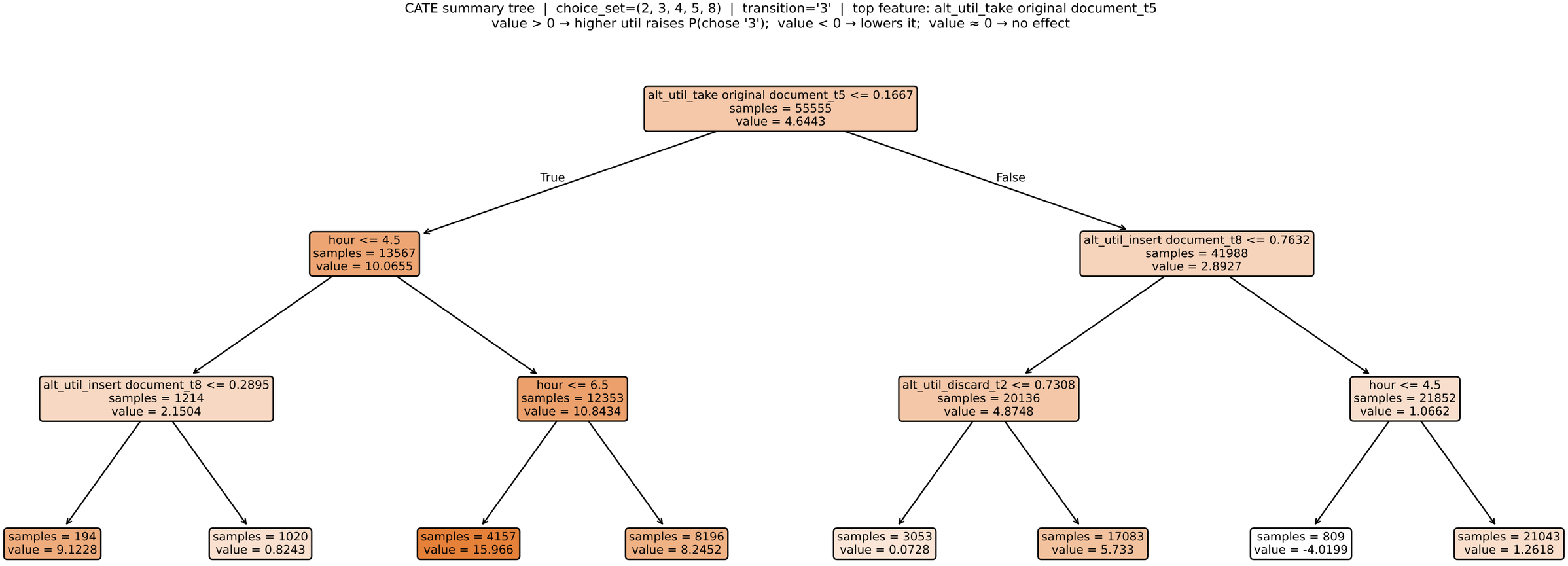

CATE summary tree | choice_set=(2, 3, 4, 5, 8) | transition='2' | top feature: alt_util_finish editing_t3
value > 0 → higher util raises P(chose '2'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 5.5
samples = 55555
value = -0.0087
True
False
alt_util_insert document_t8 <= 0.7632
samples = 8135
value = -0.0626
hour <= 7.5
samples = 47420
value = 0.0006
dayofweek <= 3.5
samples = 5379
value = 0.0144
alt_util_finish editing_t3 <= 0.7
samples = 2756
value = -0.213
alt_util_finish editing_t3 <= 0.681
samples = 11043
value = -0.012
alt_util_insert document_t8 <= 0.7947
samples = 36377
value = 0.0044
samples = 5153
value = -0.0034
samples = 226
value = 0.4198
samples = 920
value = -0.6991
samples = 1836
value = 0.0306
samples = 4476
value = 0.0076
samples = 6567
value = -0.0253
samples = 29366
value = 0.0002
samples = 7011
value = 0.0217

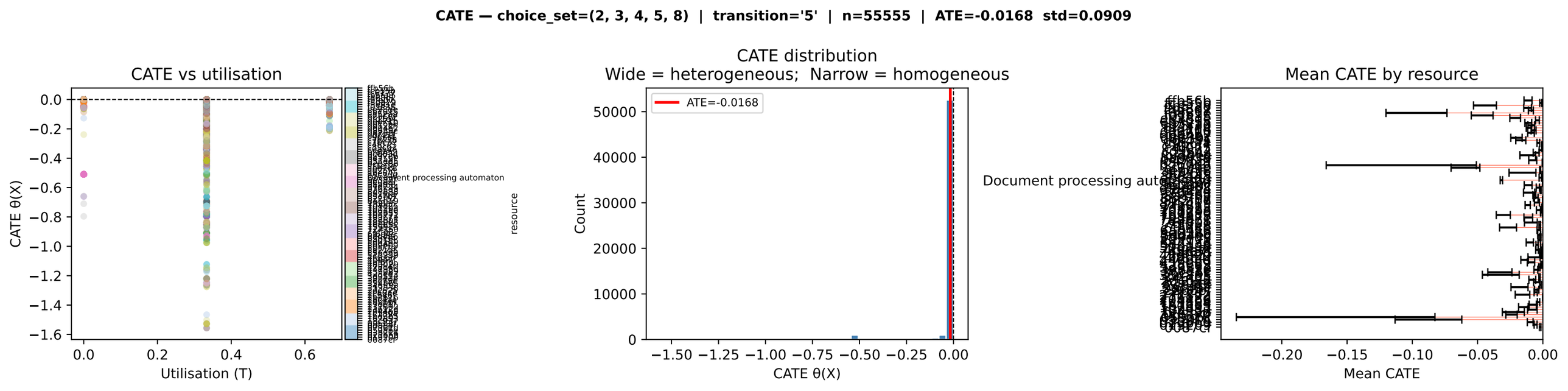
CATE — choice_set=(2, 3, 4, 5, 8) | transition='3' | n=55555 | ATE=4.6443 std=6.9191

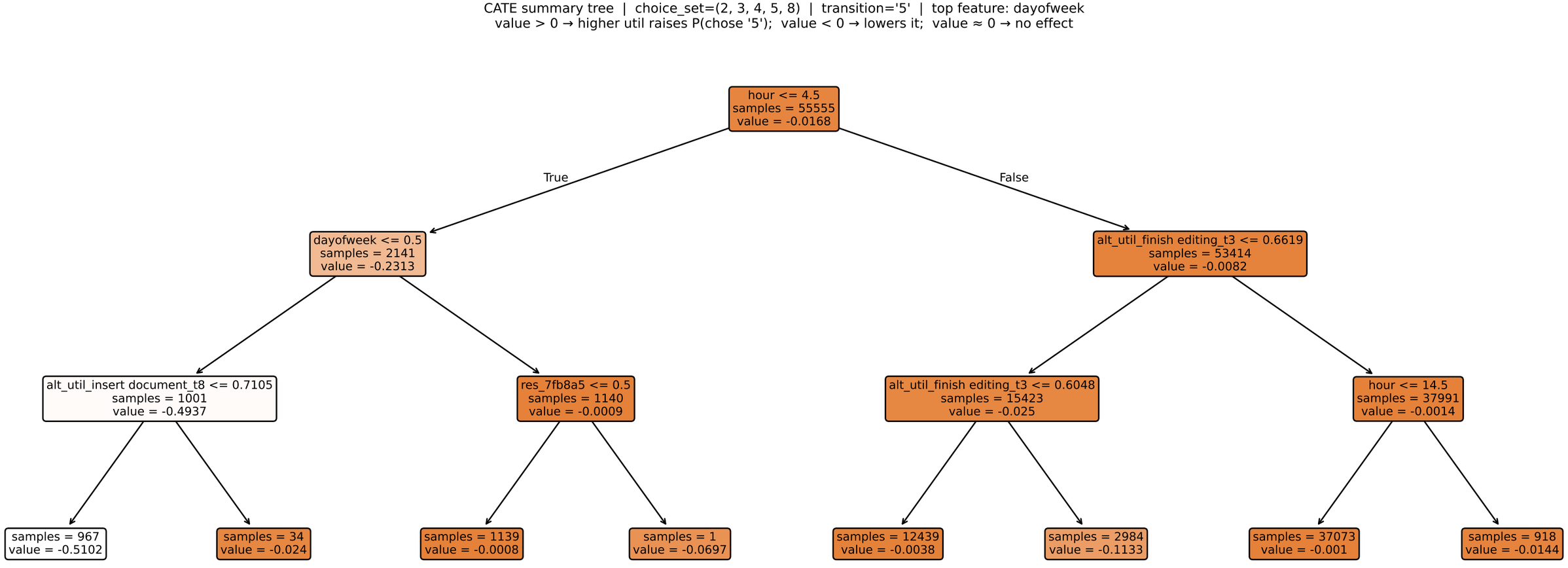


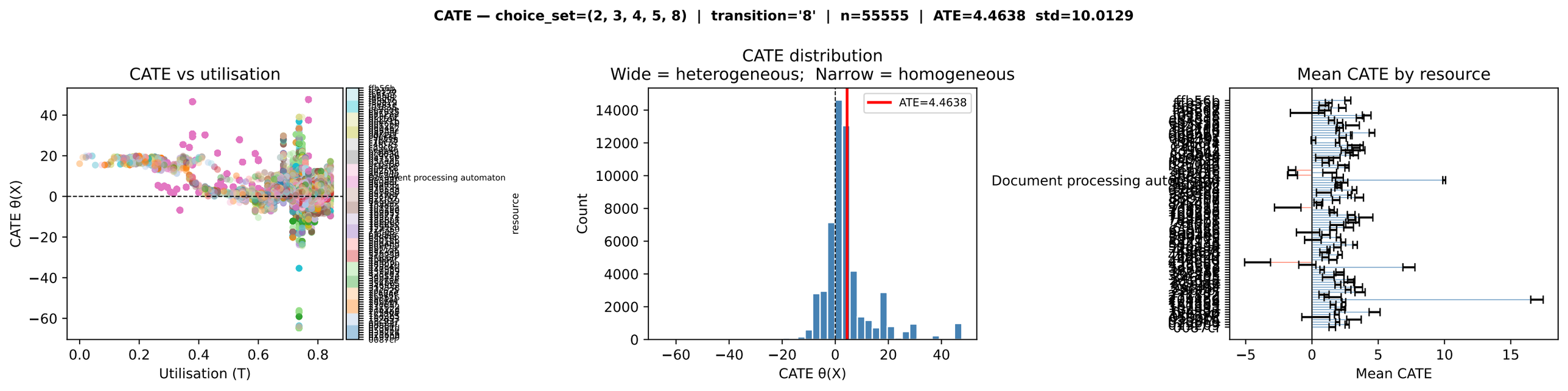





CATE summary tree | choice_set=(2, 3, 4, 5, 8) | transition='3' | top feature: alt_util_take original document_t5
value > 0 → higher util raises P(chose '3'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_take original document_t5 <= 0.1667
samples = 55555
value = 4.6443
True
False
hour <= 4.5
samples = 13567
value = 10.0655
alt_util_insert document_t8 <= 0.7632
samples = 41988
value = 2.8927
alt_util_insert document_t8 <= 0.2895
samples = 1214
value = 2.1504
hour <= 6.5
samples = 12353
value = 10.8434
alt_util_discard_t2 <= 0.7308
samples = 20136
value = 4.8748
hour <= 4.5
samples = 21852
value = 1.0662
samples = 194
value = 9.1228
samples = 1020
value = 0.8243
samples = 4157
value = 15.966
samples = 8196
value = 8.2452
samples = 3053
value = 0.0728
samples = 17083
value = 5.733
samples = 809
value = -4.0199
samples = 21043
value = 1.2618


CATE — choice_set=(2, 3, 4, 5, 8) | transition='5' | n=55555 | ATE=-0.0168 std=0.0909
CATE vs utilisation
CATE θ(X)
Utilisation (T)
resource
CATE distribution
Wide = heterogeneous; Narrow = homogeneous
ATE=-0.0168
Count
CATE θ(X)
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(2, 3, 4, 5, 8) | transition='5' | top feature: dayofweek
value > 0 → higher util raises P(chose '5'); value < 0 → lowers it; value ≈ 0 → no effect
hour <= 4.5
samples = 55555
value = -0.0168
True
False
dayofweek <= 0.5
samples = 2141
value = -0.2313
alt_util_finish editing_t3 <= 0.6619
samples = 53414
value = -0.0082
alt_util_insert document_t8 <= 0.7105
samples = 1001
value = -0.4937
res_7fb8a5 <= 0.5
samples = 1140
value = -0.0009
alt_util_finish editing_t3 <= 0.6048
samples = 15423
value = -0.025
hour <= 14.5
samples = 37991
value = -0.0014
samples = 967
value = -0.5102
samples = 34
value = -0.024
samples = 1139
value = -0.0008
samples = 1
value = -0.0697
samples = 12439
value = -0.0038
samples = 2984
value = -0.1133
samples = 37073
value = -0.001
samples = 918
value = -0.0144


CATE — choice_set=(2, 3, 4, 5, 8) | transition='8' | n=55555 | ATE=4.4638 std=10.0129
CATE vs utilisation
CATE θ(X)
Utilisation (T)
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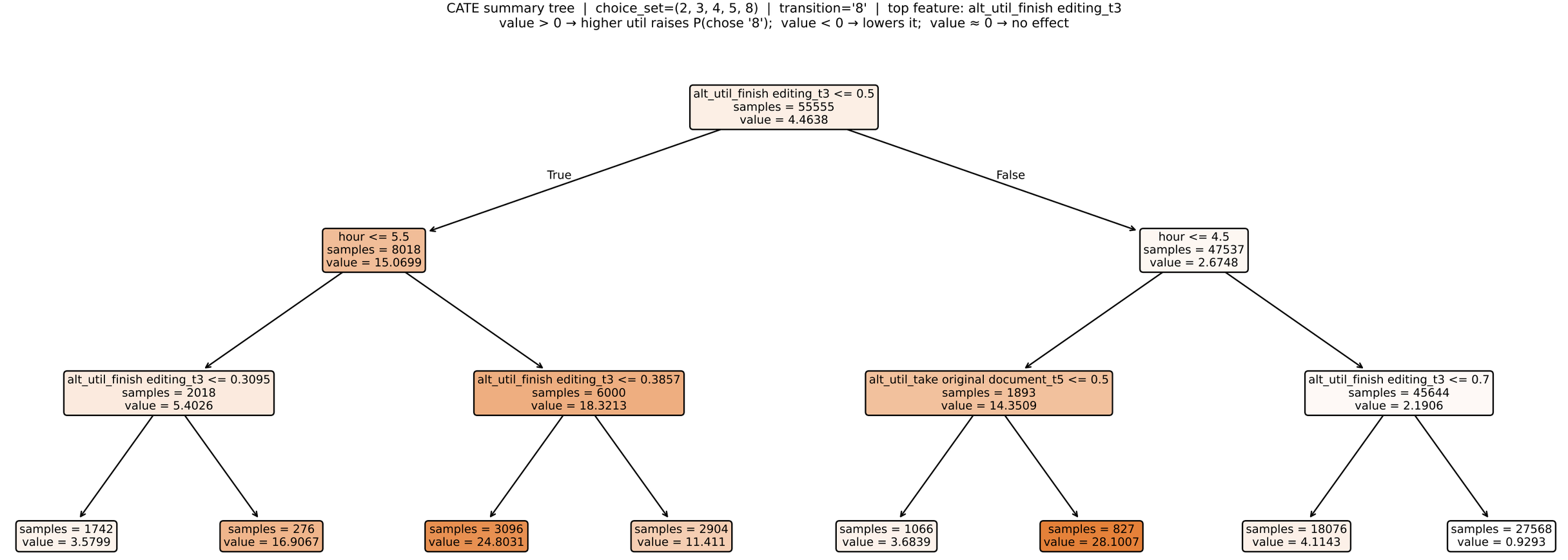
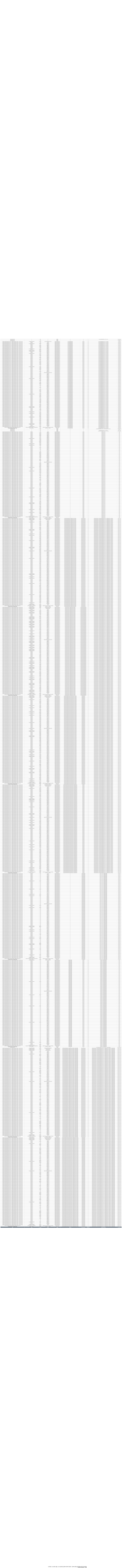
Mean CATE by resource
Mean CATE


CATE summary tree | choice_set=(2, 3, 4, 5, 8) | transition='8' | top feature: alt_util_finish editing_t3
value > 0 → higher util raises P(chose '8'); value < 0 → lowers it; value ≈ 0 → no effect
alt_util_finish editing_t3 <= 0.5
samples = 55555
value = 4.4638
True
False
hour <= 5.5
samples = 8018
value = 15.0699
hour <= 4.5
samples = 47537
value = 2.6748
alt_util_finish editing_t3 <= 0.3095
samples = 2018
value = 5.4026
alt_util_finish editing_t3 <= 0.3857
samples = 6000
value = 18.3213
alt_util_take original document_t5 <= 0.5
samples = 1893
value = 14.3509
alt_util_finish editing_t3 <= 0.7
samples = 45644
value = 2.1906
samples = 1742
value = 3.5799
samples = 276
value = 16.9067
samples = 3096
value = 24.8031
samples = 2904
value = 11.411
samples = 1066
value = 3.6839
samples = 827
value = 28.1007
samples = 18076
value = 4.1143
samples = 27568
value = 0.9293



CATE statistics summary
cate_std = heterogeneity; pct_positive/negative = share of cases where util helps/hurts; top_feature = main moderator

choice_set	transition	n	ate	cate_std	cate_min	cate_p25	cate_p50	cate_p75	cate_max	pct_positive	pct_negative	top_feature
(1, 2, 3, 4, 5, 8)	1	14750	0.217271	1.790674	-1.689626	-0.423736	-0.019748	0.303882	15.031762	47.2	52.8	alt_util_finish editing_t3
(1, 2, 3, 4, 5, 8)	8	14750	1.286132	10.725264	-29.024062	-1.418839	-0.004685	1.165476	58.792406	48.1	51.9	hour
(2, 3, 4, 5, 8)	2	55555	-0.008693	0.156651	-1.616608	-0.000337	0.001139	0.005558	1.532425	67.5	32.5	alt_util_finish editing_t3
(2, 3, 4, 5, 8)	3	55555	4.644344	6.919126	-22.009265	0.616966	3.656725	7.612583	37.387439	80.5	19.5	alt_util_take original document_t5
(2, 3, 4, 5, 8)	5	55555	-0.016784	0.090882	-1.556674	-0.0	-0.0	0.0	0.0	46.3	53.7	dayofweek
(2, 3, 4, 5, 8)	8	55555	4.463757	10.012933	-64.783464	-0.134014	2.548881	5.530872	47.701752	74.1	25.9	alt_util_finish editing_t3

Key Assumptions & Interpretation Notes

▸ Backdoor criterion (all methods)

X = {hour, dayofweek, n_alternatives, res_*} must block all confounding paths T ← C → Y.

Any variable that affects both util and routing must be in X. Unobserved confounders

cannot be detected and will bias all estimates.

▸ SUTVA (all methods)

One case's routing does not affect another's treatment or outcome.

Violated if resource occupation by one case changes the util seen by a concurrent case.

▸ Positivity (all methods)

Every covariate combination X must have positive probability for all util levels T.

Diagnosed by the covariate balance (SMD) in the backdoor check.

▸ Linearity (Double ML only)

θ is the best-fit linear slope. If the effect varies with X, θ is a variance-weighted

average of CATEs — valid as a summary, but not the simple mean.

Double ML is more powerful than the forest for testing 'is there any effect?'

▸ Honest splitting (Causal Forest only)

Tree structure and leaf estimates use separate subsamples. Yields valid CIs but

halves effective sample size → forest CIs are wider than Double ML's.

A significant Double ML θ with an insignificant forest ATE is not a contradiction.

▸ What the analysis cannot establish

Causal direction (assumes util → choice, not the reverse).

External validity beyond the observed covariate distribution.

Dynamic/lagged effects across cases over time.

Full documentation: ASSUMPTIONS.md